PHENYTOIN

Narrow therapeutic index

IV max 50 mg/min — saline only

CHEMICAL NAME 5,5-diphenylimidazolidine-2,4-dione (sodium)
CLASS Hydantoin antiepileptic (sodium-channel blocker)
TRADE NAMES Epanutin · Dilantin · Phenytek · Fosphenytoin (prodrug: Cerebyx)

THERAPEUTIC USES / INDICATIONS

- Status epilepticus (second-line after benzodiazepine).
- Focal and generalised tonic-clonic seizures.
- Seizure prophylaxis after neurosurgery/head injury.
- Trigeminal neuralgia (second-line).

ADMINISTRATION

- **IV:** dilute in 0.9% saline only (precipitates in glucose); in-line filter; max 50 mg/min (25 mg/min elderly/cardiac) with ECG and BP monitoring.
- Flush with saline before and after; large vein (purple glove syndrome).
- **PO** with or after food at the same time; suspension shaken well.
- Enteral feeds ↓ absorption — hold feeds 2 h before/after.

DOSAGES

- **Status epilepticus:** 20 mg/kg IV (max 2 g) at ≤50 mg/min.
- **Maintenance (adult):** 3–5 mg/kg/day (usually 200–400 mg/day) PO/IV.
- **Children:** loading 20 mg/kg; maintenance 4–8 mg/kg/day.
- **Target total level:** 10–20 mg/L (40–80 µmol/L); adjust for low albumin.
- Small dose changes (25–50 mg) — non-linear kinetics.

PHARMACOKINETICS

- **Onset:** IV 10–30 min. **Duration:** 24 h.
- Saturable (zero-order) liver metabolism (CYP2C9, 2C19) → small dose ↑ can cause large level ↑.
- Highly protein-bound (90%); potent enzyme inducer.
t½: 7–42 h (dose-dependent).

PHARMACODYNAMICS

- Blocks **voltage-gated sodium channels** (inactivated state) → prevents repetitive neuronal firing and seizure spread.
- Class Ib antiarrhythmic effect.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Gingival hyperplasia, hirsutism, coarse facial features, acne.
- Nystagmus, ataxia, slurred speech, dizziness (dose-related toxicity).
- Nausea, insomnia.

ADVERSE EFFECTS (SERIOUS)

- IV: hypotension, bradycardia, arrhythmias, cardiac arrest (rapid infusion).
- Purple glove syndrome, tissue necrosis.
- SJS/TEN (HLA-B*1502; Asian ancestry), DRESS.
- Blood dyscrasias, megaloblastic anaemia (folate), osteomalacia, peripheral neuropathy; fetal hydantoin syndrome.

CONTRAINDICATIONS & CAUTIONS

- IV: sinus bradycardia, SA block, 2nd/3rd-degree AV block, Adams–Stokes syndrome.
- Acute porphyria.
- Use with delavirdine.
- **Caution:** pregnancy (teratogenic), hepatic impairment, elderly.

KEY INTERACTIONS

- Enzyme inducer: ↓ effect of oral contraceptives, warfarin (variable), steroids, many antiepileptics, antiretrovirals, ciclosporin.
- ↑ **phenytoin:** fluconazole, amiodarone, isoniazid, valproate (displacement), chloramphenicol.
- **Enteral feeds, antacids:** ↓ absorption.

NURSING MANAGEMENT

- IV: continuous ECG, BP; rate ≤50 mg/min; check site frequently.
- Monitor drug levels (trough), albumin, LFTs, CBC.
- Watch for nystagmus, ataxia, confusion (toxicity).
- Seizure chart; seizure precautions.
- Oral hygiene for gum overgrowth; folate supplementation.

PATIENT EDUCATION

- Take at the same time daily; do not switch brands.
- Good dental care; regular dentist visits.
- Use non-hormonal contraception; plan pregnancy (folic acid 5 mg).
- Report rash, unsteadiness, double vision, sore throat.

KEY POINTS

- Na^+-channel blocker; non-linear kinetics.
- Level 10–20 mg/L; toxicity: nystagmus, ataxia.
- Status: 20 mg/kg IV ≤50 mg/min in saline.
- Gingival hyperplasia; enzyme inducer.